

Detecting Opinion Leaders and Trends in Online Communities

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Abstract - Today, online communities in the World Wide Web become increasingly interactive and networked. Web 2.0 technologies provide a multitude of platforms, such as blogs, wikis, and forums where for example consumers can disseminate data about products and manufacturers. This data provides an abundance of information on personal experiences and opinions which are extremely relevant for companies and sales organizations. Subjects of postings can be partly retrieved by state of the art text mining techniques. A much more challenging task is to detect factors influencing the evolvement of opinions within the social network. For such a kind of trend scouting you have to take into account the relationships among the community members. Social network analysis helps to explain social behavior of linked persons by providing quantitative measures of social interactions. A new approach based on social network analysis is presented, which allows detecting opinion leaders and opinion trends. This leads a better understanding of opinion formation. The overall concept based on text mining and social network analysis is introduced. An example is given which illustrates the analysis process.

Keywords: *Web 2.0; Network Mining; Relationship Mining; Trend Scouting*

I. INTRODUCTION

The Internet is constantly developing from a static to a highly interactive medium. Users now have the possibility of not only obtaining information but also of actively generating contents. The transformation to Web 2.0 is favored by a number of technical, economical, and social factors [7]. The increasing speed plus the decreasing costs of data transmission rates allow for development and copious usage of interactive Web applications. The new self-awareness of the information society has lead to the fact that more and more users connect online in social networks in order to exchange opinions. They interact with each other and influence their opinions reciprocally. Consumer feedback on a company's product is essential to recognize chances and risks and to implement appropriate marketing measures.

The theory of diffusion describes the spread of product opinions in social systems. According to this theory, not only the product characteristics (e.g. complexity) are an important factor in the formation of opinions but also the structure of the social networks [13]. The structure consists

of various communication relationships among the users in social networks. Who communicates with whom thus influences the making of opinions. In order to assess opinion formation in online social networks, it is necessary to analyze specific users' opinions and their communication relationships. A manual analysis is labor intensive and only possible on a limited scale. Due to the myriad of communication platforms and users in Web 2.0 it is also a great challenge. Here, an approach is introduced which automatically extracts opinions and communication patterns in forums by text mining and identifies opinion leaders and trends using social network analysis.

II. RELATED WORK

Text mining and social network analysis are currently becoming popular topics of research. Text mining allows for detecting interesting parts in texts [9] and encompasses the following subfields: information retrieval, text classification, text clustering, and information extraction [16]. While prior research focused on the discovery of facts, current work centers on emotions and opinions [9]. There are already many relevant papers on text mining in connection with opinions in Web 2.0. For example, Dave et al. [6], Liu et al. [10] as well as Popescu and Etzioni [11] describe systems which analyze customer opinions and product reviews. Here, single opinions and reviews are determined and then aggregated. However, communication processes and relationships among customers who play significant roles in opinion forming have not been taken into consideration in previous work. Social network analysis enables the examination of patterns in the relationships among interacting users [15]. It consists of an individual analysis which looks at single users' habits and a collective analysis which studies the activity structure of the entire network. Social network analysis is used increasingly for Web 2.0. The papers of Welser et al. [17], Chang et al. [4] and Gomez et al. [8] for instance explore roles, positions, and patterns of communications in online conversations. However, the conversations' content is not considered. An integrated analysis of communication content and relationships is missing.

III. APPROACH

A new approach is presented which initially detects opinions and relationships among forum users by text mining. On this basis main influential factors for opinion

forming in virtual communities are extracted. By social network analysis opinion leaders are identified and opinion involvement is analyzed. The results reveal trends in the dynamics of online social networks. The analysis of online communities comprises four steps (see Figure 1). In the first step the users' opinions on the product are extracted by text mining. In the second step, communication relationships among users are identified by text based relationship mining methods. The extracted users, opinions, and relationships form a social network which is represented as graph. Nodes and edges of the graph can be characterized by

attributes. The nodes represent the users of a forum and the edges their communication relationships. The nodes' attributes describe the users' opinions e.g. on a specific product and the edges' attributes show the frequency of communication. The resulting graph is analyzed by determining key figures for the positions of single users and for the overall structure of the network. In this way opinion leaders and opinion trends can be identified. The approach is illustrated by looking at sample forums in which opinions on Apple's iPhone® are exchanged.

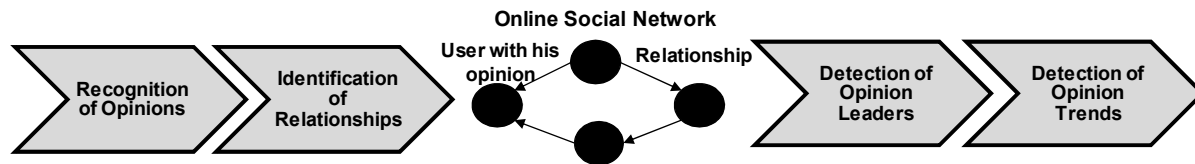


Figure 1. Research Approach

IV. RECOGNITION OF OPINIONS

The goal of opinion recognition is to detect the attitude of a user towards a product based on his/her statements in forum postings. Attitudes can generally be characterized by their polarity (positive vs. negative) and their intensity (high vs. low) [6]. The intensity of attitudes is difficult to measure in an objective manner. Polarity analysis delivers more reliable data and can be supported by text mining systems to a high degree. Polarity is modeled by three classes which are 'positive', 'neutral', and 'negative'. The automated recognition of positive, negative, and neutral opinions using text mining comprises two phases, i.e. the extraction of features from the text and the application of a learning algorithm to identify the polarity of the extracted features.

Generally, the extraction of features can be carried out by linguistic and statistical analysis [16]. After removal of unimportant stop words (e.g. 'article') the remaining words are reduced to their stem and then their frequency is calculated. Those word stems which are especially distinctive for each of the three polarity classes (meaning they appear often in one class but rarely in the others) are used as the main features of the postings. Supervised and unsupervised learning methods can be used to identify the polarity of the postings. Supervised learning describes a process in which the algorithm uses sample data to learn how to identify the polarity of a posting by its features. Unsupervised learning, in contrast, means that the algorithm uses previously determined patterns to identify the polarity.

Here, supervised learning is applied because of its enhanced learning function. In this case, the classification can be considered as the learning task. The forum postings are arranged in the three classes 'positive', 'negative', and 'neutral' based on the frequency of distinct word stems.

Different methods such as Hidden Markov Models or Maximum Entropy can be used for classification [16]. Here, support vector machines [5] are employed because of the ability to process a high number of features. Input consists of sample data records which in turn comprise various forum postings, their features, and the matching classes. By analyzing sample data the support vector machines learn the parameters of the rule which classify the postings best. The rule allows for a clustering of the postings into two groups (binary classification). In case of three classes, three rules need to be learned: 'positive' versus 'not positive', 'negative' versus 'not negative', and finally 'neutral' versus 'not neutral'. A posting will be matched to the class which has the highest probability. The rule can be illustrated as a straight line in the simple two-dimensional case. Those postings lying on one side of the line belong to the first class and those lying on the other side belong to the second class. Figure 2 shows how a straight line divides the data set on the basis of the appearance of the word stems 'love' and 'great' into the classes 'positive' and 'not positive'. After classification, the opinions are assigned to the forum users. If a user has diverging opinions he is linked to the class reflecting his strongest opinion.

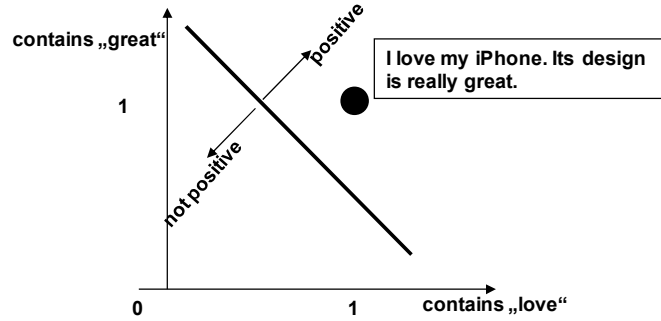


Figure 2. Classification of opinions

V. IDENTIFICATION OF RELATIONSHIPS

A relationship between two forum users exists if one user (sender) refers to another user (recipient) in at least one posting. A communication relationship is characterized by the sender, the recipient, and the communication frequency.

In the field of text mining there are two approaches which deal with the recognition of relationships among entities in texts: co-reference resolution and relationship extraction [16]. The aim of co-reference resolution is to recognize if two words in one text refer to the same object. A typical application is to determine whether a specific pronoun refers to a previously mentioned noun. For example, in the statement ‘The iPhone is great. It has a huge display,’ it must be determined whether the word ‘iPhone’ and ‘it’ refer to the same object. The relationship extraction deals with the identification of different relationships

between two components of a sentence. For example, the task could be to find the relationship ‘is producer of’. In the phrase ‘Apple’s iPhone is great,’ it must be detected that Apple is the producer of the iPhone. Looking abstractly at this challenge, the goal is to solve a binary classification problem. Each pair of possible text entities will be divided into two classes: ‘has defined relation’ and ‘has no defined relation’. The classification is based on statistical-linguistic attributes of the one or the other text element and their common attributes. Again, supervised as well as unsupervised learning can be applied for classification.

When classifying communication relationships of users in a forum, each posting of a user is analyzed for relationships with previous postings of different users. Figure 3 depicts all possible pairs out of three postings and their classification.

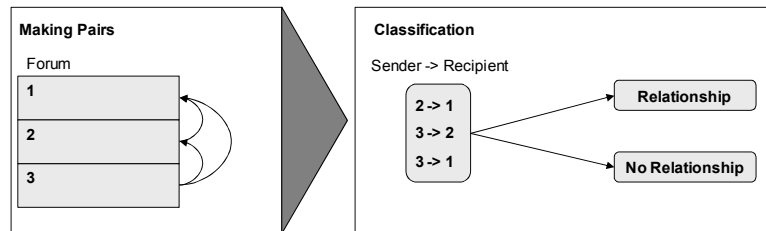


Figure 3. Classification of Communication Relationships

Important signs indicating the existence of a relationship between two users are:

- mentioning of the recipient’s name in the sender’s posting
- quotations of the recipient’s posting in the sender’s posting
- distance of the postings in the forum
- appearance of specific words in both postings

These criteria are checked by string matching and used as attributes for classification. The classification is done by support vector machines. After classification, the relation frequency of all pairs is calculated by summation.

V. DETECTION OF OPINION LEADERS

The extracted users with their opinions and relationships form a social network which is analyzed first on user level. The goal is to identify opinion leaders. Opinion leaders are persons who play a crucial role in forming opinions. Their special position and their communication habits within the network allow them to influence the opinion of many other users [2]. They offer an orientation for users who are searching information. The identification of opinion leaders provides great potential for marketing. The orientation function of opinion leaders with positive opinions can be used to deliberately spread specific information. Opinion leaders with negative views can be counteracted in a targeted way.

There are many key indicators in the field of social network analysis which describe the position and the

communication habits of users. When observing opinion leaders, the key indicator “centrality” is of special importance. It marks on a scale from 0 to 1 how central a user is within the network [14]. Three different figures are common: degree centrality, closeness centrality, and betweenness centrality. Degree centrality measures how frequently a user communicates directly with others. It is calculated as the fraction of the sum of all outgoing and incoming relationships of one user and the sum of all relationships among all users in the network [15]. Closeness centrality measures the closeness of one user to all other users. It is defined as the inverse sum of all path distances from the observed user to all others [15]. Betweenness centrality shows how often a specific user can be found on the shortest connecting path of all pairs of users. The betweenness centrality of a user is calculated as the fraction of the number of the shortest paths the user is lying on and the number of all shortest paths [15].

Degree centrality is an indicator for local opinion leaders. Forum users with high degree centrality communicate very frequently directly with other users and

have the possibility of influencing them. Their influence, however, is limited to their direct environment. Closeness and betweenness centrality not only look at the direct communication relationships but also at indirect ones and thus are indicators for global opinion leaders. Forum users who have the shortest distance to all other users are most likely to be noticed by others. They are very independent from the information transferred by other users. Those users found on communication paths between other users have the highest chance of controlling the information flow between users.

Figure 4 shows an exemplary thread from the forum computerbase.de. User 28 has the highest degree centrality with a value of 0.13. However, the user’s positive opinion on the iPhone is only shared by his closer environment. User 34 with a closeness and betweenness centrality of 0.3 is closest to all other users and controls the information flow best. He has a negative opinion which is also with 49% the most frequent opinion class in the forum (only 34% positive opinions and 17% neutral opinions).

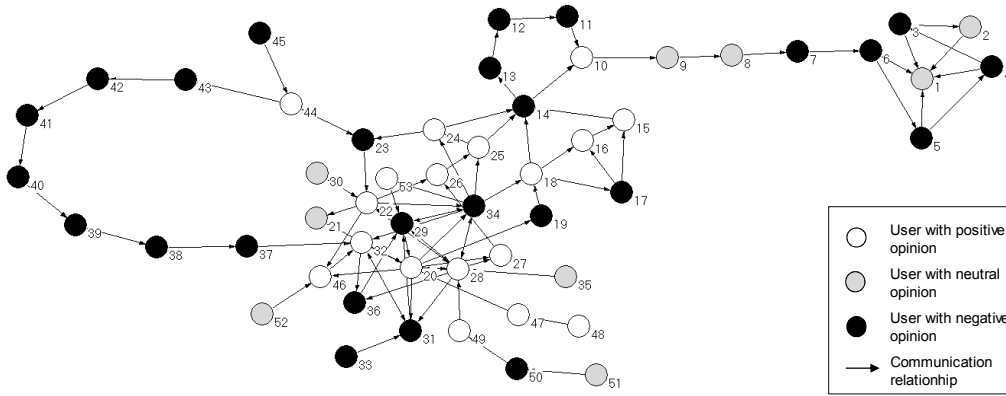


Figure 4. Thread from the forum computerbase.de

For the first validation, in this case 17 threads from forums regarding the iPhone are chosen. The focus is put on opinions of the users with the highest value of closeness and betweenness centrality as well as on the overall opinion within the forum in terms of the most frequent opinion class. The validation is carried out via Spearmansch correlation coefficient which measures direction and intensity of the correlation between ordinary scaled variables on a scale from -1 (perfect negative correlation) to 1 (perfect positive correlation)

[1]. A value of 0 shows independence between both variables. In social sciences, values higher than 0.3 are considered as important correlations [3]. The Spearmansch correlation coefficient of 0.56 shows an important relation between the opinion of the user with the highest closeness centrality and the overall opinion in the forum. The positive correlation between the opinion of the user with the highest betweenness centrality and the overall opinion was at a value of 0.49 and significant as well.

VI. DETECTION OF OPINION TRENDS

The aim of the analysis on network level is to explain and forecast the formation of opinions in a forum. Two aspects of opinion formation are of special interest: first, the characteristics of a network which lead to a particular opinion trend and second, the characteristics of a network in which the trend follows the opinion of the opinion leader. An early identification of positive and negative trends allows to plan and support effective reinforcing and counteracting measures.

Social network analysis provides a number of key figures which describe the structure of the entire network. For analyzing opinions, the key figures density, Randic connectivity, and closeness centralization are especially relevant. Density measures the connection of a network and is an indicator for communication within the network. It is calculated as the ratio of the number of existing communication relationships to the number of all possible relationships [14]. The higher the calculated value of density the more information can be exchanged within the network.

Randic connectivity shows the level of branching within the network. It is the sum of the inverse roots of the multiplied node degrees of all user pairs [12]. A low value indicates a high level of branching which means that many users have direct relationships to a high number of users. Closeness centralization indicates the variation of the closeness centrality values of the network's users. It is defined as the ratio of the sum of the deviations of all users' closeness centrality from the closeness centrality of the most central user to the theoretically possible centralization [15]. A network is strongly centralized if there are only a few central and many peripheral users.

Density and Randic connectivity indicators show if there is a trend of the overall forum opinion. The higher the density and the lower the Randic connectivity, the more forum users are connected with each other and communicate directly. In this way, opinions can be exchanged increasingly. Thus, there is a high probability of a trend of the opinion of all users regarding a specific product. The trend becomes stronger the more one opinion class outweighs the others. It is calculated as the product of the deviations of the largest opinion class from the other two.

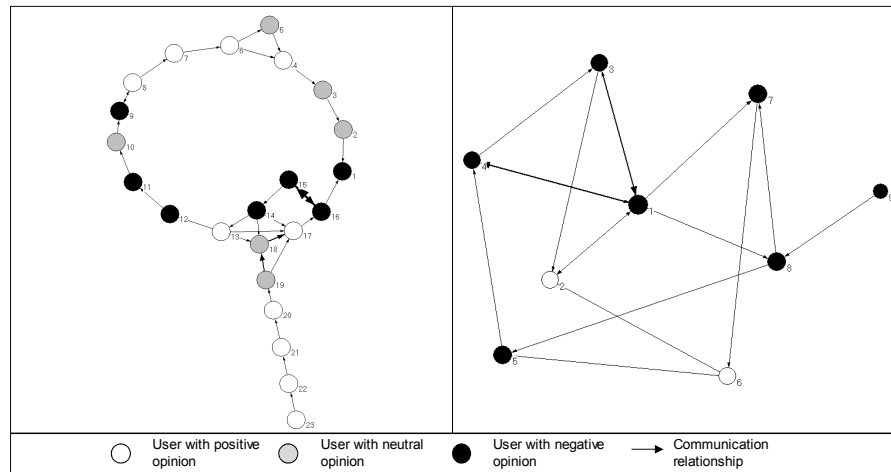


Figure 5. Threads from the forums slashfm.de and pcfreunde.de

Figure 5 shows two threads regarding the iPhone. The thread from the forum slashfm.de (Figure 5, left hand side) has a relatively low density of 0.08 and a relatively high Randic connectivity of 11.33. It also shows a low opinion trend of 0.03. 44% of the users have a positive, 30% have a negative, and 26% a neutral opinion on the iPhone. The thread from pcfreunde.de (Figure 5, right hand side), in contrast, has a relatively high density of 0.22 and a low Randic connectivity of 4.22. The opinion trend is relatively high with a value of 0.43. 78% of the users think negatively about the iPhone and 22% think positively.

Centralization indicates whether a trend arises from an opinion leader. If there is a high value of closeness centralization, there is one or a few opinion leaders in the center of the network and many users on the periphery. The opinion of the leader can thus be spread easily. If the network is, in addition, strongly linked the leader's opinion can be spread among many other forum users which could start a trend following the leader's opinion. The pcfreunde.de thread in Figure 5 illustrates this. The closeness centralization is relatively high with a value of 0.4. 78% of users share the negative opinion of the opinion leader (user 1) who has the highest closeness centrality of 0.7.

The first validation was based on 17 threads about the iPhone. The key figures density, Randic connectivity, closeness centralization, opinion of the opinion leader, and

tendency of the overall opinion were calculated. The coherence was then measured by using Pearson's correlation coefficient and Cramer's contingency coefficient. The Pearson correlation coefficient enables a measurement of intensity and direction of correlations between quantitative variables on a scale from -1 (perfect negative correlation) to 1 (perfect positive correlation)

[1]. A value of 0 indicates that there is no correlation at all. Cramer's contingency coefficient shows the intensity of the correlation between two nominally scaled variables on a scale from 0 (no correlation) to 1 (perfect correlation)

[1]. The correlation coefficient shows an important positive correlation between density and tendency with a value of 0.54. There is also a significant, positive correlation between Randic connectivity and tendency with a value of 0.50. Cramer's contingency coefficient of 0.55 shows a significant, positive correlation between high density, low Randic connectivity, and high centralization as well as a trend evoked by the opinion leader.

VII. CONCLUSIONS

The Internet is changing more and more from a mere source of information to an interactive platform. Customers form social networks where they discuss products and influence each other's opinions. For companies, it is very important to learn about opinions on their products in order to assess chances and risks. A manual analysis is only

feasible on a very limited scale. An automated computer supported analysis is necessary given the large number of virtual communities with huge amounts of postings. A new approach for analyzing opinions in forums enables an automated extraction of opinions and relationships from forums and identification of opinion leaders and opinion trends. The approach is realized by combining text mining and social network analysis. Thus existing positive opinions can be reinforced via marketing measures. Negative opinions can be counteracted in time e.g. by product improvements or appropriate marketing measures. The approach can be considered as a basic concept for analyzing opinions in social networks. A future task is the expansion of the validation database which should also include forums on various product categories. Moreover, a dynamical network analysis will be added to the static social network examination. Snapshots of social networks at different, consecutive points in time will be taken. Thus, opinion changers can be identified on user level and opinion trends can be detected on network level.

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